

Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

Research Interests: Autonomous Systems, Planning and Control, and Multi-Robot SLAM

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Education

Carnegie Mellon University (CMU) Master of Science in Robotics (MSR)	Aug. 2026 – Expected 2028 Pittsburgh, PA
University of Michigan, Ann Arbor (UM) B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science	Oct. 2024 – May 2026 GPA: 3.85/4.0 (Top 5%)
• Core Coursework: Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization	
Shanghai Jiao Tong University (SJTU) B.S.E. in Electrical and Computer Engineering	Sep. 2022 – Jul. 2026 GPA: 3.56/4.0 (Top 20%)
• Core Coursework: Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems	

Research Experiences and Internships

Intelligent Robotics and Autonomy Lab , UM Research Assistant, Directed by Prof. Vasileios Tzoumas	Oct. 2024 – Apr. 2026 Ann Arbor, MI
• Multi-Robot Simulation and ROS2: Rebuilt a Unity + AirSim multi-drone testbed with revised dynamics and sensing; migrated SlideSLAM communication from ROS1 to ROS2 for bandwidth-limited active-SLAM evaluation.	
• Communication-Aware Exploration: Developed distributed EXP3-style algorithms using submodular objectives and bandit feedback to balance map coverage and communication cost in simulation.	
Computational Autonomy and Robotics Lab, UM Research Assistant, Directed by Prof. Maani Ghaffari	Jun. 2025 – Apr. 2026 Ann Arbor, MI
• Learning-Augmented Lie-MPC: Developed and field-tested an autonomous surface vehicle controller that estimates residual wind/current disturbances online and corrects the dynamics used by MPC. Achieved a 20% improvement in trajectory-tracking performance over baseline MPC in real-ASV experiments.	
• Field Perception: Built DINO/SAM2 segmentation and fine-tuned Grounded-DINO/YOLO for submerged vegetation detection in autonomy experiments.	
• Sensor Integration: Designed mounts and integrated, aligned, and calibrated GPS, lidar, IMU, and cameras for field data collection and closed-loop control.	
Unitree Robotics, R&D Department, Embodied AI Team Research Intern, Humanoid Robot Foundation Models	May 2026 – Present Hangzhou, China
• Real-Robot Policy Infrastructure: Built and validated a closed-loop stack on Unitree H2/G1 humanoids integrating teleoperation, VLA/WAM and whole-body policy adapters, rollout logging, evaluation, controller interfaces, and policy fine-tuning; enabled one-command launches across platforms.	

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Graduate Student Researcher, advised by Prof. Changliu Liu

Research experience continued

Aug. 2026 – Present

Pittsburgh, PA

- **Whole-Body and Safe Locomotion Research (ongoing):** Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion.
- **Dexmate Bimanual Manipulation Demo, IROS 2026:** Contributed to a bimanual block-stacking demonstration combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration.

Shanghai Artificial Intelligence Lab & IRMV, SJTU

Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang

Apr. 2026 – Present

Shanghai, China

- **Iterative Visual KV Compression:** Studied errors from reusing retained visual KV entries after neighboring tokens are removed in multi-turn and streaming inference. Developed context-aware re-encoding at original positions and compared with inheritance and restoration under matched memory and runtime budgets.

Publications and Manuscripts

- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari. “Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control”, *IEEE Robotics and Automation Letters (RA-L)*, revised manuscript under review, 2025.
- Xiaohan Wang^{*}, **Ziyu Xu**^{*}, Xuyi Yang. “Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”, *Submitted to ICLR 2027*. ^{*}*Equal contribution*.
- Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and **Ziyu Xu**. “StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assistants”, *Submitted to ICLR 2027*.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban. “Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, *npj Digital Surgery*, accepted, 2024.

Selected Awards

Gold Medal, University Physics Competition (2023); Dean’s Honor List, University of Michigan (2024, 2025).

Skills

- **Autonomy:** Active-SLAM simulation, communication-aware exploration, MPC, RL, perception, sensor integration and calibration
- **Tools:** Python/C++, ROS/ROS2, Unity + AirSim, PyTorch, OpenCV, Linux, Docker, Git, CAD